

Procedural Discourse Annotation Framework

for Agent Traces — Coauthor Contribution Report · Version 3.0

June 2026 · HPOP ICLR Paper

Version 3.0 — What changed

- ▶ Complete 5-step pipeline: Steps 1a → 1b → 2 → 3 → 4 → 5 all implemented and validated on T1–T5
- ▶ CPA layer (Step 1b): 13 Canonical Procedural Actions, 4 rounds stress-test, ABSTAIN=0%
- ▶ Skill-Type Classification (Step 2): phase-sequence matching against SKILL_LIBRARY
- ▶ Dependency Labeling (Step 3): nodes = CPA instances (not phrase, not phase)
- ▶ Confidence Scoring (Step 4): HIGH/MEDIUM/ABSTAIN with label distribution monitoring
- ▶ Human Review (Step 5): rule iteration mechanism, ≥3 occurrences → new rule recommendation
- ▶ ADJACENT_ONLY reduced: 18% → 6% after B3 WebLINX patch rules (R1/R2/R3)
- ▶ Output: skills.jsonl (5 traces) + local_orders.jsonl (52 edges)

PDAF_v3_package.zip
pdaf_v3_package/
├── README.md ←
├── pipeline/
│ ├── pdaf_pipeline_v3.py ← 5 pipeline
├── theory/
│ ├── CPA_Rules_Final.md ← 13 CPA 4 stress-test
│ └── skills_schema.md ← skills.jsonl schema
├── outputs/
│ ├── skills_v3.jsonl ← HPOP 1 5 trace
│ ├── local_orders_v3.jsonl ← HPOP 2 52 edges
│ └── review/ ← T1–T5
├── traces/ ← T1–T5 trace
├── skills/ ← S1–S5 Markdown skill
└── reports/
 ├── Coauthor_Report_PARI_v3_final.docx ← report
 └── Weekly_Update_*.docx

CPA	v1	action vocabulary	UI	"	"	"	"	HPOP	UI
skill									
Phrase		Phrase	HPOP						
HPOP									
3	→2	phrase_units.jsonl		Phrase		Phrase			
Phase	R	HPOP UI		HPOP	Section 5.4	formulation	phase	HPOP	
subroutine type									
Step 2		WebLINX			trace		skill		
QC	v3	LLM	Step 1b	LLM				CPA mapping	
v1									

新增 CPA层 Step 1b	action vocabulary = click / scroll / type (UI操作, 无语义区分) SEARCH ≡ FILL (无法区分)	action vocabulary = 13个CPA SEARCH_SOURCE FILL_FIELD 等	Activity Theory (Leontiev 1978): 可复用结构在Action层, 不在Operation层 HPOP需要语义vocabulary
删除 Phrase层 B1 + Step 3	phrase = local functional unit 几个action组成一个phrase dependency节点 = phrase_id phrase_units.jsonl输出	phrase层删除 dependency节点 = cpa_instance_id 直接在CPA间标注	phrase在HPOP数学模型里 没有对应的数学对象; CPA有Activity Theory依据, 节点语义更清晰
简化 输出文件 3个→2个	① skills.jsonl ② phrase_units.jsonl ③ local_orders.jsonl (节点=phrase_id)	① skills.jsonl ② local_orders.jsonl (节点=cpa_instance_id) phrase_units删除	HPOP的segment boundary是 skill-level, 不是phrase-level; phrase_units.jsonl在HPOP的 formulation中无对应结构
重新定位 Phase = R A层定位	phase = annotation标签 (给人类标注员用) skill = 更高层的概念 两者关系不明确	phase直接 = HPOP的skill-type vocabulary R 可重复出现	HPOP UI截图验证: Skill Type 列 (dialogue-clarify等) = 我们 的phase粒度; 同类型可重复 出现 (HPOP论文Section 5.4)
重写 Step 2 skill分类推断	手动指定skill_id 查SKILL_LIBRARY名字 没有推断逻辑 真实数据无法适用	phase-sequence Jaccard匹配 + domain加分项 自动推断skill	Script Theory (Schank & Abelson 1977): skill = script with typical event sequence; 真实数据需要自动推断
重定向 Step 4/5 QC范围	评估dependency label 是否可靠 裁决歧义dependency 没有覆盖CPA mapping	评估CPA mapping + dependency edge + label分布监控 规则迭代 (≥3次)	rule-guided LLM输出是概率 性的, v1的QC没有针对 LLM执行的质量控制; Settles (2009) 规则迭代

理论不变: A层 Miller & Cohen (2001) · Grosz & Sidner (1986) · Nelson & Narens (1990) · Hutchins (1995) 全部保留
新增引用: Leontiev (1978) · Sutton (1999) · Schank & Abelson (1977) · Settles (2009)

Layer A/B/C **v3**

Layer A	Neuro-cognitive Abstraction →											
"	"	9	phase	CPA	phase							
v1	A	"		"	v3	A	HPOP	skill-type vocabulary	R	9	phase	"
"	"		HPOP									
Layer B	Linguistic Formalization →										B1	B2/B3
B1	Phrase			—	phrase	v3		B2	8	dependency labels		
		phrase pair		CPA	instance pair	B3	5	ambiguity rules		3	WebLINX-	
specific	STATE_CHANGE				R1/R2/R3	B4	QC	protocol				
dependency label					CPA mapping							
Layer C	Annotation Pipeline →										6	5
	Step 3	Phrase Segmentation				Step 1	1a	1b		CPA Abstraction		Step 2
"	"	"phase-sequence	"									

1. Problem

1.1 The core gap

The HPOP model learns reusable skill structures from flat agent execution traces. To do this reliably it needs structured weak supervision: (a) an action vocabulary that encodes procedural meaning rather than UI events, and (b) skill-type labels and dependency priors to initialise its inference.

Core gap	<i>HPOP's action vocabulary A must encode what the agent is trying to accomplish, not which UI element it clicked. Without a principled CPA layer and dependency annotation, the model learns UI event co-occurrence patterns, not reusable procedural skills.</i>
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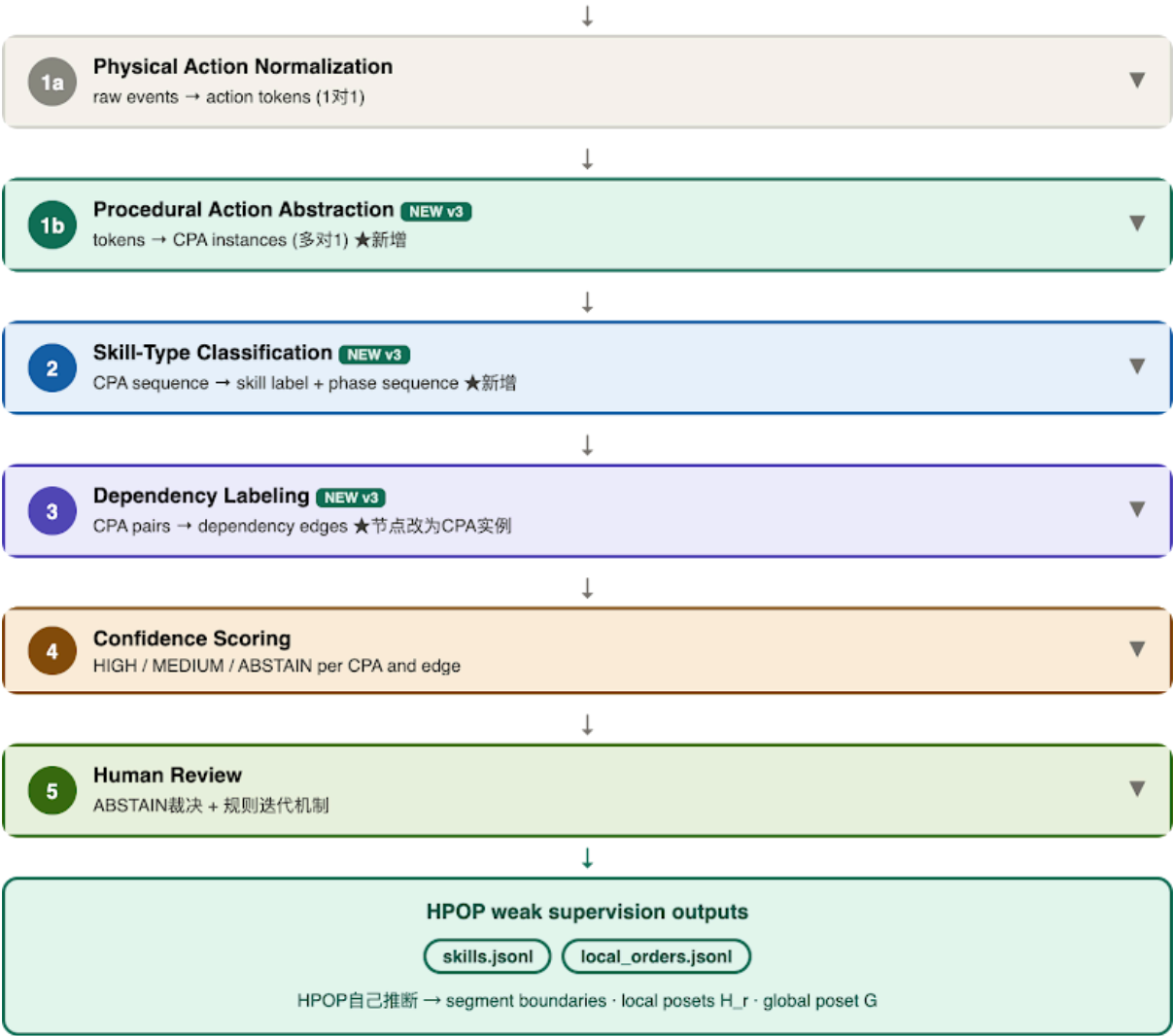
1.2 What this work provides

- Action vocabulary $A = \{1, \dots, M\}$: 13 Canonical Procedural Actions grounded in Activity Theory (Leontiev, 1978), replacing raw click/scroll/type
- Skill-type vocabulary $R = \{1, \dots, R\}$: 9 phase categories grounded in cognitive neuroscience (Miller & Cohen, 2001), serving as HPOP's subroutine type labels
- Weak supervision file 1 — skills.jsonl: skill label priors with CPA instance sequences
- Weak supervision file 2 — local_orders.jsonl: dependency edge priors between CPA instances

2. Framework

The framework has three components: the CPA layer (action vocabulary), the A-layer phase taxonomy (skill-type vocabulary), and a 5-step annotation pipeline.

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► NEW in v3.0 — entire component

2.1 CPA Layer — Action Vocabulary

2.1.1 Why click/scroll/type is insufficient

Raw agent traces consist of UI-level operations — the lowest-level observable events. Following Activity Theory (Leontiev, 1978), these are Operations: automatised, routinised executions that carry no goal-directedness. The critical problem is that identical operation sequences represent completely different procedural Actions:

Physical sequence	Context A	Context B
click → type → click	Search for a source (RETRIEVE)	Fill a login form (EXTRACT)
scroll → read	Browse search results (INSPECT — no proposition)	Verify a specific claim (VERIFY — explicit proposition)

goal-directed behaviour:

Activity Theory tier	Definition	Our framework	Example
Activity	Motivated by a goal	Skill (verify_fact)	Verify whether a claim is true
Action ★	Goal-directed, consciously executed, temporally extended	CPA (SEARCH_SOURCE etc.)	SEARCH_SOURCE · VERIFY_CLAIM
Operation	Automatised, routinised	Physical action	click · scroll · type

CPAs correspond to the Action tier. Reusable procedural structure emerges at this level (Botvinick et al., 2009; Sutton et al., 1999) — not at the Operation level. This is the principled justification for the CPA layer.

2.1.3 The 13 Canonical Procedural Actions

13 CPAs were defined by cross-referencing A-layer phase definitions against the WebLINX action distribution (Lù et al., 2024). CHECK_SOURCE was removed after stress-testing: WebLINX contains no corresponding action_type, so the rule never fired. 12 rule fixes were applied across 4 stress-test rounds on 23 constructed test traces.

Phase	CPA	Maps from	Signal	Theo
PLAN	FORMULATE_GOAL	plan/load(homepage)	position==0 Miller (2001 Botvi (2009 Hutch Hutch Nelso (1990	
RETRIEVE	SEARCH_SOURCE	type(*)+click(*) — 5 variants	keyword or Enter depe Waso Flowe (1981 PREC satisf	STAT
RETRIEVE	OPEN_DOCUMENT	click(*)+load(*)	URL artifact	
RETRIEVE	NAVIGATE_PAGE	tabcreate / tabswitch	new tab context	
INSPECT	READ_SOURCE	scroll+read / scroll alone	no proposition	
INSPECT	SCAN_RESULTS	scroll(results_keywords)	result/listing/item	
EXTRACT	EXTRACT_EVIDENCE	copy(*)	specific content	
EXTRACT	FILL_FIELD	fill_field × 1+	form field	
VERIFY	VERIFY_CLAIM	compare(proposition_keywords)	PROPOSITION_KEYWORDS	
WRITE	WRITE_RESPONSE	write(*)	new text artifact	
WRITE	SUBMIT_FORM	click(submit_keywords)	13 keywords	

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REPAIR	REPAIR_SOURCE	go_back()	failure signal	NORM Shall
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2.1.4 Stress-test results

4 rounds of stress-testing across 23 constructed test traces covering all 13 CPAs and their most common boundary cases:

Round	HIGH	MEDIUM	ABSTAIN	Key fix
Round 1 — original 14 rules	41%	14%	11%	Baseline
Round 2 — 10 fixes	62%	32%	6%	Extended keywords; scroll alone → READ_SOURCE; tabswitch
Round 3 — 11 fixes	62%	32%	6%	Window-based lookahead replaces greedy token processing
Round 4 — 12 fixes (final)	64%	36%	0% ✓	click(box)+type+press(enter) variant; click(search_btn) alone

MEDIUM (36%) is expected: scroll alone and single fill_field are genuinely ambiguous without broader context. These cases are resolved by Confidence Scoring (Step 4) and Human Review (Step 5), not forced into HIGH.

► REVISED in v3.0 — phases now directly serve as HPOP's skill-type vocabulary R

2.2 Layer A — Phase Taxonomy and Skill-Type Vocabulary

Layer A defines nine phase categories grounded in cognitive neuroscience. In Version 3.0 these phases directly constitute HPOP's skill-type vocabulary $R = \{1, \dots, R\}$. The same phase type may recur multiple times within a single trace — exactly as HPOP's subroutine recurrence model requires.

HPOP alignment	<i>In the HPOP Skill Instances UI, 'dialogue-clarify' appears 4 times in one trace (S1, S3, S6, S8). Our A-layer phase RETRIEVE plays the same role: it recurs whenever a new retrieval sub-task begins. HPOP's coarser 'Phase' column (Task Reception / Execution / Evaluation...) is HPOP's own inferred global poset — NOT something we define.</i>
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L level	Phase	CPAs	Cognitive state + reference
L1 goal-setting	PLAN	FORMULATE_GOAL	Goal decomposition — PFC sets goal hierarchy (Miller & Cohen, 2001) · intentional discourse segment (Grosz & Sidner, 1986)
L2 execution	RETRIEVE	SEARCH · OPEN · NAVIGATE	Directed subgoal pursuit — externally-directed acquisition (Botvinick et al., 2009)

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L2 execution	EXTRACT	EXTRACT_EVIDENCE · FILL	Focused subgoal pursuit — central executive locks on target (Baddeley, 2000)
L2 execution	VERIFY	VERIFY_CLAIM	Metacognitive control — explicit proposition; output is truth value (Nelson & Narens, 1990 · Wason, 1960)
L2 execution	WRITE	WRITE · SUBMIT	Generative production — externalization mode (Flower & Hayes, 1981)
L3 integration	SYNTHESIZE	SYNTHESIZE_FINDINGS	Multi-source integration — assembles representation no single unit possessed (Hutchins, 1995)
L4 interrupt	REPAIR	REPAIR_SOURCE	Error monitoring — SAS override after failure signal (Norman & Shallice, 1986)
L4 interrupt	HANDOFF	(agent identity change)	Distributed cognitive load transfer (Hutchins, 1995 · Monsell, 2003)

► REVISED in v3.0 — 6 steps → 5 steps; phrase layer removed; dependency nodes = CPA instances

2.3 Layer C — 5-Step Annotation Pipeline

The pipeline implements the annotation theory in five executable steps. Phrase Segmentation and the separate Dependency Labeling step from v2 have been restructured: dependency labeling now operates directly on CPA instance pairs (not phrase pairs), eliminating the need for a phrase segmentation layer.

Step	Name	What it does	Theory	Output
1a	Physical Normalization	Raw events → (action_type, tool_family, agent_id, artifact_id). Collapse repeated actions within 5s window.	ISO 24612:2012	Action tokens
1b ★	CPA Abstraction	Groups tokens into CPA instances via priority-ordered contextual rules + window-based lookahead. 13 CPAs, 12 rule fixes.	Leontiev (1978) · Sutton et al. (1999) · Botvinick (2009) · Wason (1960) · Norman & Shallice (1986)	CPA sequence
2 ★	Skill Classification	CPA sequence → skill label via phase-sequence matching against SKILL_LIBRARY (Jaccard score + domain bonuses).	Miller & Cohen (2001) · Grosz & Sidner (1986) · Schank & Abelson (1977)	skills.jsonl
3 ★	Dependency Labeling	CPA pairs → dependency edges. 8 labels, priority-ordered. Nodes = cpa_instance_id (not phrase, not phase).	Kildall (1973) · PDTB · Nelson & Narens (1990) · Norman & Shallice (1986)	local_orders.jsonl
4	Confidence Scoring	HIGH (≥0.8) / MEDIUM (0.5–0.8) / ABSTAIN per CPA and	Cohen (1960) κ · B4 QC protocol	Review flags

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		warning).		
5	Human Review	ABSTAIN adjudication; OPTIONAL_SAMPLE 20% random; ≥3 recurring pattern → rule update recommendation.	Settles (2009) Active Learning	Gold set + rule update

2.3.1 The 8 dependency labels

Labels are applied in strict priority order to each pair of consecutive CPA instances. The first matching rule wins. ADJACENT_ONLY is a residual fallback and should be rare in well-formed traces (v3 achieves 6%).

#	Label	Definition	Theory
1	REPAIR_OF	B exists because A failed or produced unusable output	Norman & Shallice (1986) SAS · Luo et al. (2025)
2	VERIFY_OF	B's primary function is to test whether A's output is correct	Nelson & Narens (1990) control · Wason (1960)
3	DATA_FLOW	A's artifact flows directly into B (RAW dependency)	Kildall (1973) program dependence analysis
4	STATE_CHANGE	A modifies shared state (viewport, form, working memory) that B reads	Program dependence analysis · B3 WebLINX patches R1–R3
5	PRECONDITION	A must complete before B; no artifact flows (control dependence)	Kildall (1973) control dependence
6	ELABORATES	B executes a subgoal stated or implied by A	Mann & Thompson (1988) RST · Hobbs (1979)
7	INCOMPARABLE	A and B serve the same parent goal; order is arbitrary	Workflow-specific partial-order label
8	ADJACENT_ONLY	No structural relation (residual fallback — should be rare)	Workflow-specific fallback

2.3.2 B3 WebLINX patch rules (new in v3)

Three new STATE_CHANGE rules were added based on the ADJACENT_ONLY diagnostic (v3 dropped ADJACENT_ONLY from 18% to 6%):

- R1: SEARCH_SOURCE → READ_SOURCE: search loads results page, changing viewport state; reading depends on this state
- R2: READ_SOURCE → OPEN_DOCUMENT: reading results reveals which links exist; document opening depends on link-visibility state
- R3: READ_SOURCE → FILL_FIELD: reading the page reveals form structure; field filling depends on page state

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skills.jsonl — HPOP skill label priors

One line per trace. Records the skill name, CPA instance sequence, and derived phase sequence. The `global_phase_sequence` field may contain repeated phase labels (e.g. RETRIEVE twice in one trace) — this matches HPOP's subroutine recurrence design.

```
{
  "trace_id": "T5",
  "skill_id": "S005",
  "skill_name": "verify_fact",
  "skill_file": "S5_verify_fact.md",
  "cpa_instances": [
    {"cpa_instance_id": "T5-CPA1", "cpa": "FORMULATE_GOAL", "phase": "PLAN",
     "turn_start": 1, "turn_end": 1, "artifact_id": "google.com", "confidence": 0.90},
    {"cpa_instance_id": "T5-CPA4", "cpa": "EXTRACT_EVIDENCE", "phase": "EXTRACT",
     "turn_start": 5, "turn_end": 5, "artifact_id": "claim_A_google", "confidence": 0.92},
    ...
  ],
  "global_cpa_sequence": ["FORMULATE_GOAL", "SEARCH_SOURCE", "READ_SOURCE", ...],
  "global_phase_sequence": ["PLAN", "RETRIEVE", "INSPECT", "EXTRACT", "RETRIEVE", "INSPECT", "EXTRACT", "VERIFY", "WRITE"]
}
```

local_orders.jsonl — HPOP local partial order priors

One line per dependency edge. Nodes are `cpa_instance_id` — not `phrase_id`, not `phase_id`. This directly provides HPOP's local poset `Hr` with CPA-level action ordering constraints.

```
{
  "edge_id": "T5-E13",
  "trace_id": "T5",
  "skill_name": "verify_fact",
  "source_cpa_id": "T5-CPA4",
  "target_cpa_id": "T5-CPA10",
  "source_cpa": "EXTRACT_EVIDENCE",
  "target_cpa": "VERIFY_CLAIM",
  "label": "VERIFY_OF",
  "rationale": "extracted evidence flows into verification (Wason 1960)",
  "confidence": 0.90,
  "artifact_transferred": "claim_A_google"
}
```

3.1 Pipeline validation — T1–T5

The complete 5-step pipeline was validated on 5 constructed traces (T1–T5) covering all 13 CPAs and all 8 dependency labels.

Trace	Skill	CPAs	HIGH	ABSTAIN	Edges	ADJ_ONLY
T1	search_and_summarize	7	6 (86%)	0	9	0 (0%)
T2	compare_products	11	7 (64%)	2	10	1 (10%)
T3	fill_and_submit_form	6	2 (33%)	2	4	1 (25%)
T4	search_and_retrieve	9	7 (78%)	0	13	0 (0%)
T5	verify_fact	11	8 (73%)	0	16	1 (6%)
Total	5 skills	44	30 (68%)	4 (9%)	52	3 (6%)

Label	Count	Interpretation
DATA_FLOW	10	Artifact transfer between CPAs — most fundamental dependency type
STATE_CHANGE	16	Viewport/memory/form state modifications — largest category, correctly captures most web agent dependencies
PRECONDITION	8	Control dependencies (tab context, goal gating)
REPAIR_OF	4	Failure-and-repair patterns — T4 shows complete go_back→retry cycle
ELABORATES	4	Goal decomposition from FORMULATE_GOAL
INCOMPARABLE	4	Parallel sub-tasks (e.g. two RETRIEVE instances in T5)
VERIFY_OF	3	Evidence-to-verification edges in T5
ADJACENT_ONLY	3 (6%)	Residual fallback — within acceptable range (threshold: 15%)

3.2 Key findings from T5 (verify_fact)

T5 is the most theoretically important trace because it exercises INCOMPARABLE (parallel retrieval), VERIFY_OF (evidence-to-verification), and the INSPECT vs VERIFY distinction simultaneously.

Step	Content
Step 1a	13 raw events → 13 action tokens
Step 1b	11 CPAs (0 ABSTAIN): FORMULATE_GOAL → SEARCH_SOURCE → READ_SOURCE → EXTRACT_EVIDENCE → NAVIGATE_PAGE → [3×READ_SOURCE] → EXTRACT_EVIDENCE → VERIFY_CLAIM → WRITE_RESPONSE
Step 2	Skill: verify_fact (S005). global_phase_sequence: [PLAN, RETRIEVE, INSPECT, EXTRACT, RETRIEVE, INSPECT, INSPECT, INSPECT, EXTRACT, VERIFY, WRITE] — RETRIEVE and INSPECT both recur

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Step 4	4 flagged: 0 mandatory, 3 optional (MEDIUM READ_SOURCE instances), 1 advisory (1 ADJACENT_ONLY)
Step 5	0 rule updates triggered (no pattern reached ≥ 3 threshold in this trace)

4. Impact

4.1 Direct contribution to HPOP

The framework provides HPOP with two vocabularies it cannot derive from raw click/scroll/type sequences:

- Action vocabulary $A = \{1, \dots, M\}$: the 13 CPAs, produced by Step 1b. HPOP uses this to represent each trace as a CPA sequence rather than a UI event sequence — enabling discovery of reusable skill patterns.
- Skill-type vocabulary $R = \{1, \dots, R\}$: the 9 A-layer phases. HPOP uses this as the label space for its subroutine types, with `global_phase_sequence` in `skills.jsonl` as the initialisation prior.
- Local partial order priors: `local_orders.jsonl` provides CPA-level dependency edges for HPOP's `H_r`. HPOP's own inference then refines these into the final local and global posets.

4.2 Package deliverables

- `pdaf_pipeline_v3.py`: complete 5-step pipeline, ~500 lines, standard library only, Python 3.10+
- `skills_v3.jsonl`: 5 traces, skill labels + CPA instance sequences + phase sequences
- `local_orders_v3.jsonl`: 52 dependency edges with rationale and confidence scores
- T1–T5 review JSONs: human review queues with mandatory/optional/advisory tiers
- `CPA_Rules_Final.md`: 13 CPA rules with priority ordering and 4-round stress-test log
- T1–T5 trace files + S1–S5 Markdown skill files
- `Coauthor_Report_PARI_v3.docx`: this report

4.3 Open questions for supervisor

◆ REQUIRES SUPERVISOR INPUT

#	Question	Why it matters
1	Are the 9 A-layer phases confirmed as HPOP's skill-type vocabulary R?	Determines whether A-layer is used as-is or needs a separate skill layer
2	Is <code>local_orders.jsonl</code> (CPA-level dependency edges) the correct format for HPOP's local partial order priors?	Determines whether nodes should be CPA instances, or coarser
3	Is Step 1b confirmed as rule-guided LLM (vs pure rule-based)?	Determines whether Confidence Scoring / Human Review remain core steps

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INPUT

Raw agent trace — flat event log

e.g. click(search_box) · type('query') · scroll(down) · copy(text)

↓

1a

Physical Action Normalization

raw events → action tokens (1对1)

输入

raw events

click(search_box) · type('query') · go_back()

输出

action tokens

(action_type, tool_family, agent_id, artifact_id)

执行内容

解析每个raw event的action_type
(click/scroll/type/load/copy...)
根据TOOL_FAMILY_MAP推断tool_family
(browser/keyboard/clipboard/reader)
保留agent_id和artifact_id字段
去除5秒内重复的相同动作 (去噪)

理论依据

ISO 24612:2012 (LAF)
annotation unit的数据结构标准
Bird & Liberman (1999)
Annotation Graphs多层标注框架

1对1映射, 每个event对应一个token, 不做聚合

1b

Procedural Action Abstraction

NEW v3

tokens → CPA instances (多对1) ★新增

输入

action tokens

[click, type, click, scroll, read, copy, write]

输出

CPA sequence

[SEARCH_SOURCE, READ_SOURCE, EXTRACT_EVIDENCE, WRITE_RESPONSE]

执行内容

Window-based lookahead (同时看当前+后1+后2个token)
按优先级顺序匹配13条CPA规则 (第一个匹配获胜)
4轮stress-test (23个traces): ABSTAIN从11%降到0%
每个CPA打confidence: HIGH(≥0.8) / MEDIUM(0.5-0.8) / ABSTAIN(0)

理论依据

Leontiev (1978) Activity Theory
CPA = Action层 (goal-directed); click = Operation层 (automatised)
Sutton et al. (1999) Options
CPA = temporally extended goal-directed action
Botvinick et al. (2009)
RETRIEVE类CPA的hierarchical subgoal pursuit
Wason (1960)
VERIFY_CLAIM的命题判定规则
Norman & Shallice (1986)
REPAIR_SOURCE的失败信号触发

13个CPA

FORMULATE_GOAL SEARCH_SOURCE OPEN_DOCUMENT NAVIGATE_PAGE READ_SOURCE SCAN_RESULTS EXTRACT_EVIDENCE FILL_FIELD VERIFY_CLAIM WRITE_RESPONSE SUBMIT_FORM SYNTHESIZE_FINDINGS REPAIR_SOURCE

13个CPA = HPOP的action vocabulary A = {1,...,M}

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输入

CPA sequence

[FORMULATE_GOAL, SEARCH_SOURCE, EXTRACT_EVIDENCE, VERIFY_CLAIM, WRITE_RESPONSE]

输出

skills.jsonl

skill_name · global_cpa_sequence · global_phase_sequence

执行内容

从CPA推断每个action对应的phase (CPA_TO_PHASE映射表)
用observed phase_sequence与SKILL_LIBRARY做Jaccard-style匹配推断skill
加分项: VERIFY出现→verify_fact分高 / REPAIR出现→search_and_retrieve分高
同一个phase可以在一条trace里出现多次 (HPOP的 subroutine recurrence)

理论依据

Miller & Cohen (2001)
9个phase = PFC认知控制状态
Grosz & Sidner (1986)
phase = intentional discourse segment
Schank & Abelson (1977) Script Theory
skill = script, phase_sequence = 典型事件序列
HPOP paper (2026)
同一个subroutine type可在一条trace里多次出现

9个PHASE

PLAN RETRIEVE INSPECT EXTRACT VERIFY WRITE SYNTHESIZE REPAIR HANDOFF

9个phase = HPOP的skill-type vocabulary $R = \{1, \dots, R\}$

3

Dependency Labeling NEW v3

CPA pairs → dependency edges ★节点改为CPA实例

输入

CPA instances

T5-CPA4 (EXTRACT_EVIDENCE) · T5-CPA10 (VERIFY_CLAIM)

输出

local_orders.jsonl

edge_id · source_cpa_id · target_cpa_id · label · rationale

执行内容

节点: cpa_instance_id (不是phrase, 不是phase)
对每对相邻CPA pair按B2优先级顺序标dependency (第一个匹配获胜)
同时检查相邻(i,i+1)和隔一个(i,i+2)——处理non-adjacent PRECONDITION
v3新增B3 WebLINX patches R1/R2/R3: ADJACENT_ONLY
从18%降到6%

理论依据

Kildall (1973)
DATA_FLOW(RAW) · STATE_CHANGE · PRECONDITION(控制依赖)
Nelson & Narens (1990) · Wason (1960)
VERIFY_OF的metacognitive control
Norman & Shallice (1986)
REPAIR_OF的失败信号
Mann & Thompson (1988) RST · Hobbs (1979)
ELABORATES标签

8个DEPENDENCY LABELS

REPAIR_OF VERIFY_OF DATA_FLOW STATE_CHANGE PRECONDITION ELABORATES INCOMPARABLE ADJACENT_ONLY

52 edges跨T1-T5 · ADJACENT_ONLY=3(6%) · STATE_CHANGE最多(16)

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References

★ Green left border = new in Version 3.0

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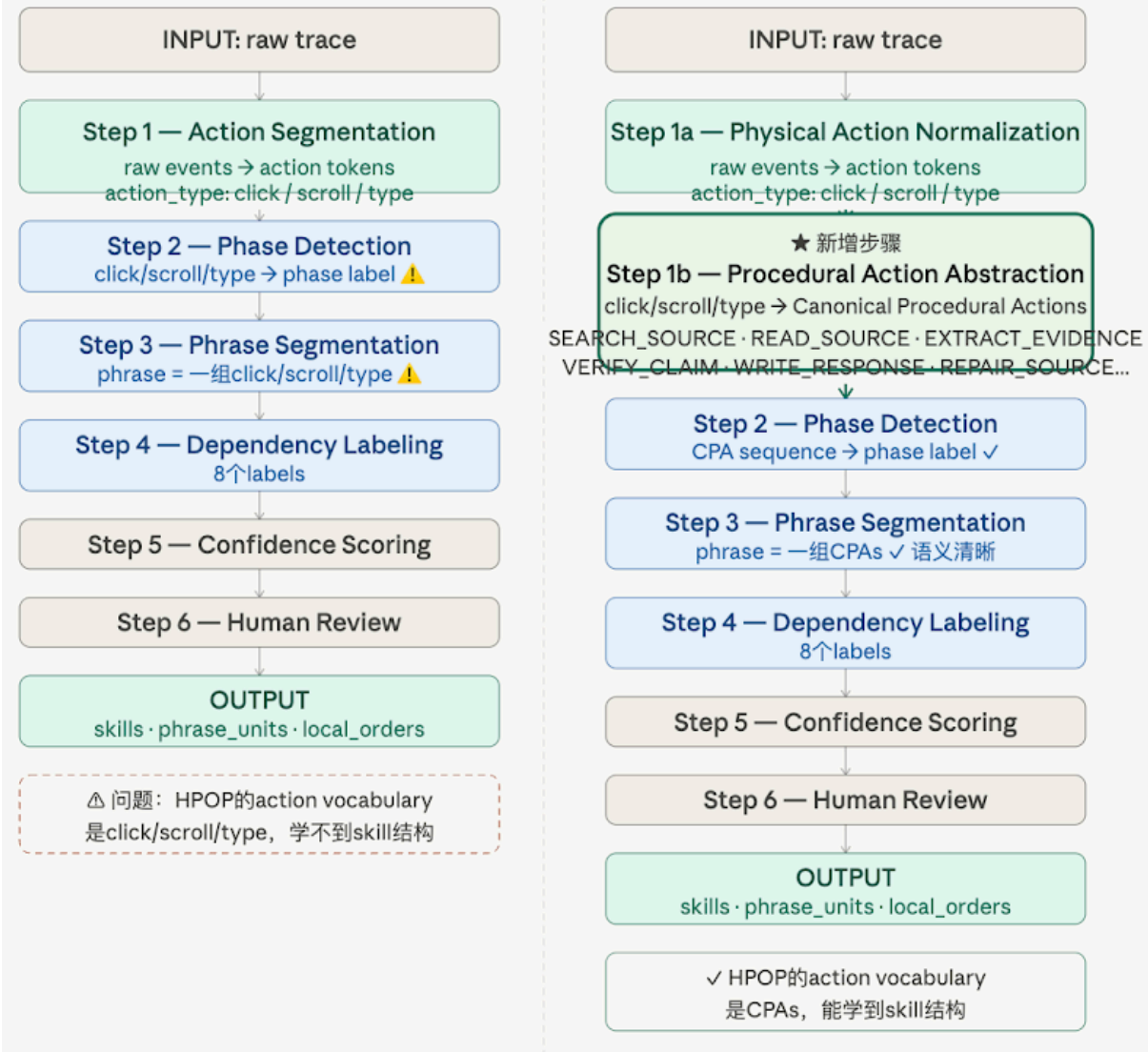
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Previous Adaptation:

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Step 1b "click → type → click" HPOP
click/scroll/type RETRIEVE EXTRACT UI workflow mining procedural skill
discovery Step 1b CPA Leontiev 1978 Activity Theory Step 1b
14 CPA HPOP SEARCH_SOURCE → READ_SOURCE → EXTRACT_EVIDENCE
skill pattern Step 1b CPA.

Step 1b — Procedural Action Abstraction Activity Theory

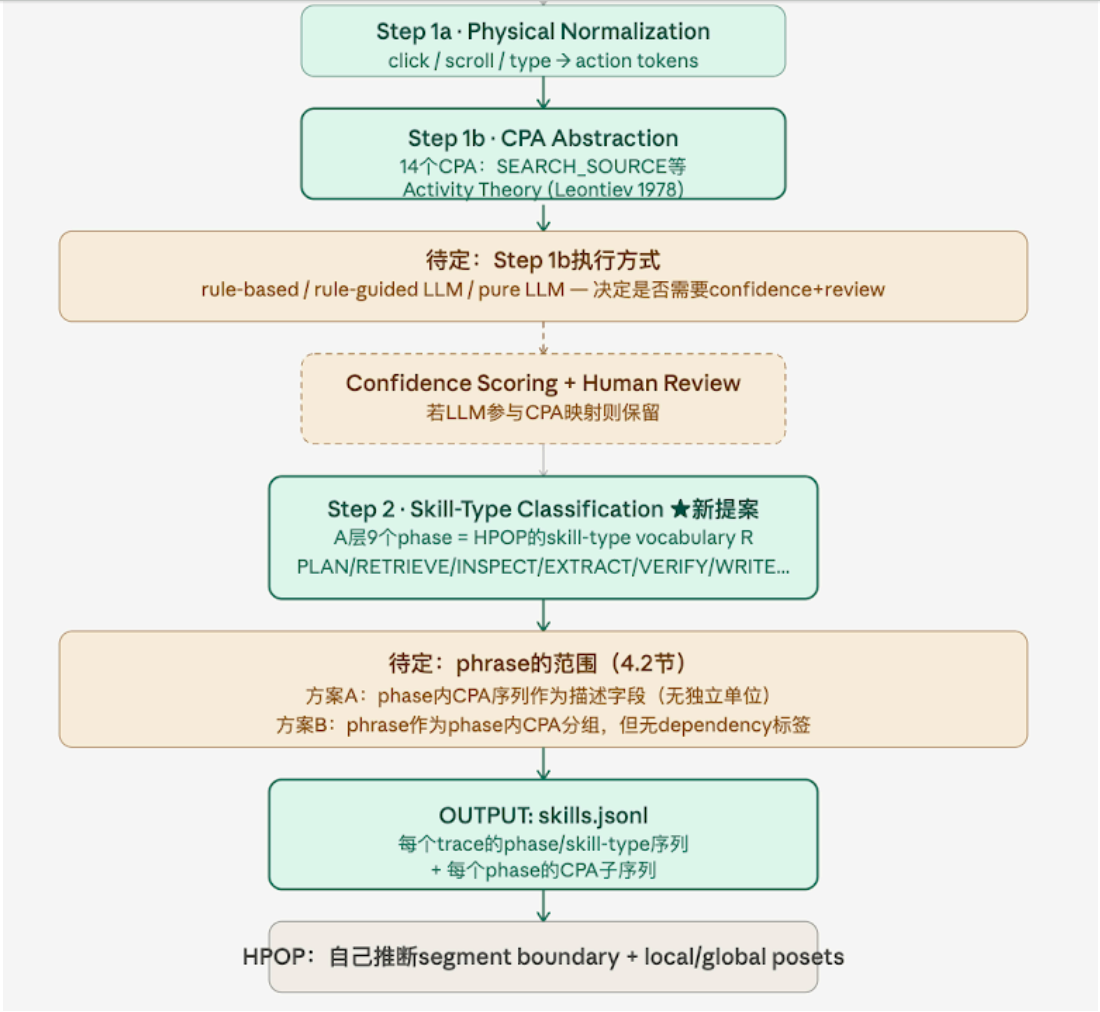
Leontiev 1978 Activity Theory

Activity → HPOP skill
Action → CPA SEARCH_SOURCE ← Step 1b
Operation → click/scroll/type

skill structure Action Operation HPOP CPA
click/scroll/type

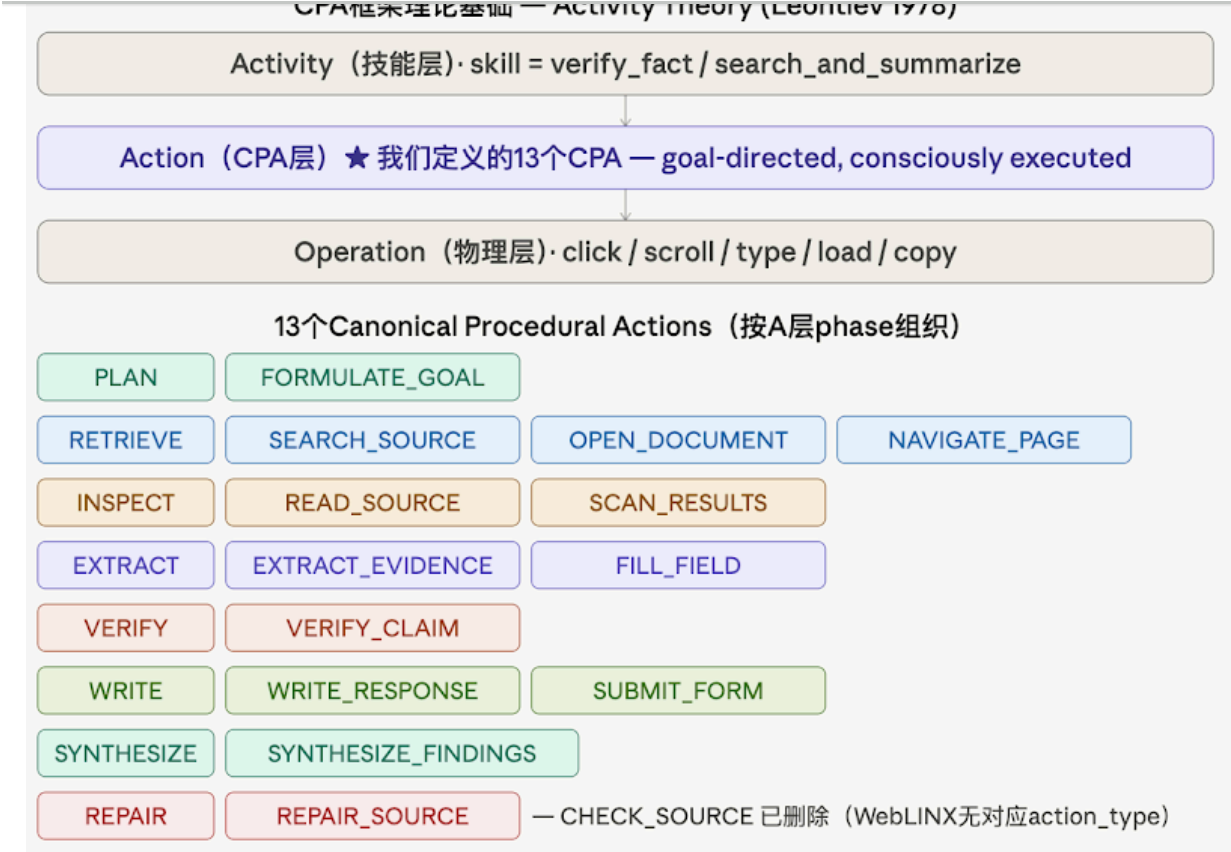
Version 2 (Simplified Adjusted Version) Steps 1B +Step 2 Skill-type Classification

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Details in Canonical Procedural Actions			CPA	CPA	— Activity Theory		
CPA							
UI-level actions	HPOP		13(14)	Canonical Procedural Actions (CPA Step 1a/1b)			
physical action		contextual signal					
CPA		Activity Theory	Leontiev, 1978				
Activity	click/scroll/type						
Action	SEARCH_SOURCE/VERIFY_CLAIM						
Operation	verify_fact			cognition	9	phase	
	canonical procedural actions		raw trace	9	phase		L1 —
goal-setting	L2 — execution	L3 — integration	L4 — interrupt				
HPOP	skill structure						

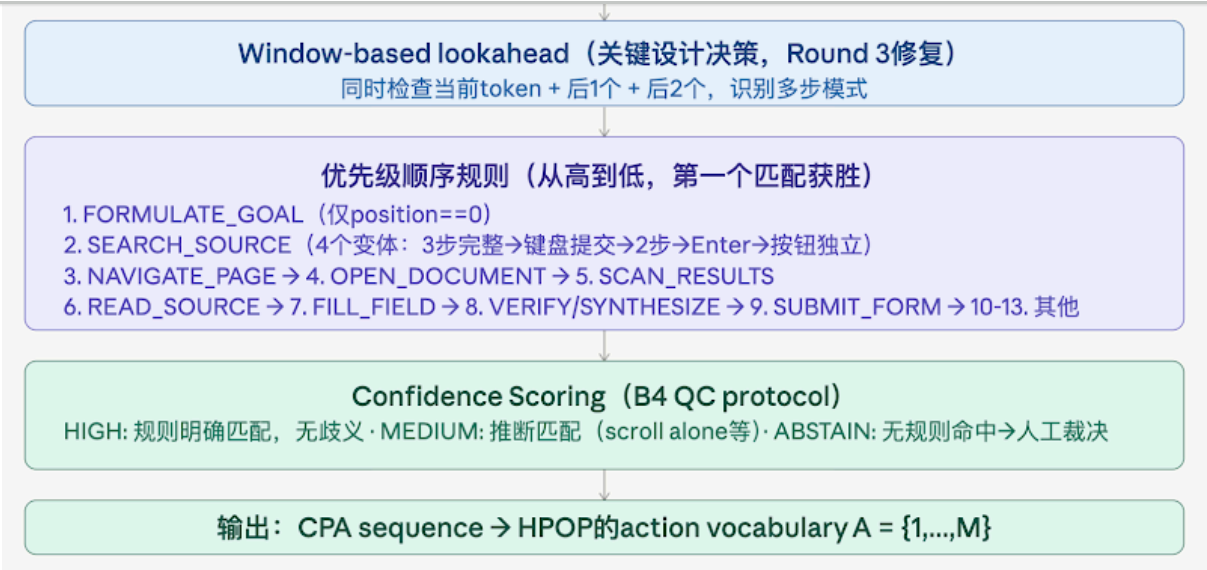
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13 CPA

CPA	A	phase	phase	phase	CPA
CPA					
		physical actions			
		phase	phase detection		
		WebLINX			
*CHECK_SOURCE	ABSTAIN		——WebLINX raw trace	"check"	action_type

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window-based lookahead

"click(search_box) + type + press(enter)"

ABSTAIN

window-based

greedy

token

click

token

click

search_button

token

token

SEARCH_SOURCE

OPEN_DOCUMENT

click

token

OPEN_DOCUMENT

"click(search_box)+type+click"

keyword

SCAN_RESULTS

results

/listing/item/product/hit/search/found

SUBMIT_FORM

"results"

"submit"

confirm/continue/next/book/pay

四轮stress-test结果 (23个test traces)				
轮次	HIGH	MEDIUM	ABSTAIN	修复数
Round 1 (原始14条规则)	41%	14%	11% △	0
Round 2	62%	32%	6% □	10
Round 3 (window-based lookahead)	62%	32%	6% □	11
Round 4 (最终)	64%	36%	0% ✓	12

MEDIUM 36% 属于正常范围 (scroll alone / single fill_field需要上下文推断)

stress-test

CPA 14 13 CHECK_SOURCE WebLINX "check" action_type

← Report 3

SUBMIT_FORM "submit" 13 confirm continue next book pay

greedy token window-based lookahead click

MEDIUM 36% scroll alone single fill_field
LLM HIGH

ABSTAIN 11% 0% MEDIUM 14% 36%

ABSTAIN case MEDIUM— 100%
MEDIUM Confidence Scoring LLM
READ_SOURCE MEDIUM ABSTAIN 99% scroll alone
READ_SOURCE scroll+read scroll alone

36% MEDIUM Confidence Scoring Human Review Step 1b rule-
guided LLM LLM MEDIUM HIGH

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HPOP看到的是：
SEARCH · OPEN · READ · EXTRACT · WRITE
SEARCH · OPEN · SEARCH · OPEN · COMPARE · WRITE
(两条trace, CPA都有重叠)

HPOP需要同时判断三件事：

- ① 哪里是skill边界？
- ② 每个CPA属于哪个skill？
- ③ CPA之间是什么依赖关系？

搜索空间太大
HPOP容易陷入degenerate solution

CPA sequence:
SEARCH · OPEN · READ · EXTRACT · WRITE
(5个CPAs)

Phrase层 (我们的贡献)

P1: [SEARCH-OPEN] = "找到目标页面"
P2: [READ-EXTRACT] = "读取并提取内容"
P3: [WRITE] = "写出回应"

Skill: search_and_summarize

P1 → (DATA_FLOW) → P2 → (DATA_FLOW) → P3

- ✓ 问题1 (局部): phrase内部CPA依赖
 - ✓ 问题2 (全局): phrase间skill结构
- 两个问题分开解决, 搜索空间大幅缩小

HPOP的数学模型为什么需要phrase层

HPOP有两层偏序结构：

local poset Hr: skill内部的action偏序

global poset G: skill类型之间的偏序

local poset里的"action"必须是phrase级别 — 太细 (click) 没有语义, 太粗 (skill) 和自身同级

类比：语言学的层级分析

语言学：词 → 短语(phrase) → 句子 → 话语

不会直接：词 → 话语 (缺少中间的组织层级)

我们的框架：CPA → phrase → skill

不直接：CPA → skill (phrase是必要的中间组织单位, Grosz & Sidner 1986)

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A 9 phase —

10

CPA 13 CPA —

Activity Theory	CPA	Sutton	Options Framework
CPA	"temporally extended actions"	Wason 1960	Norman & Shallice 1986
	VERIFY_CLAIM REPAIR_SOURCE		
14	keyword lists	SUBMIT_KEYWORDS	13
testing"	ICLR	"validated through WebLINX stress-	

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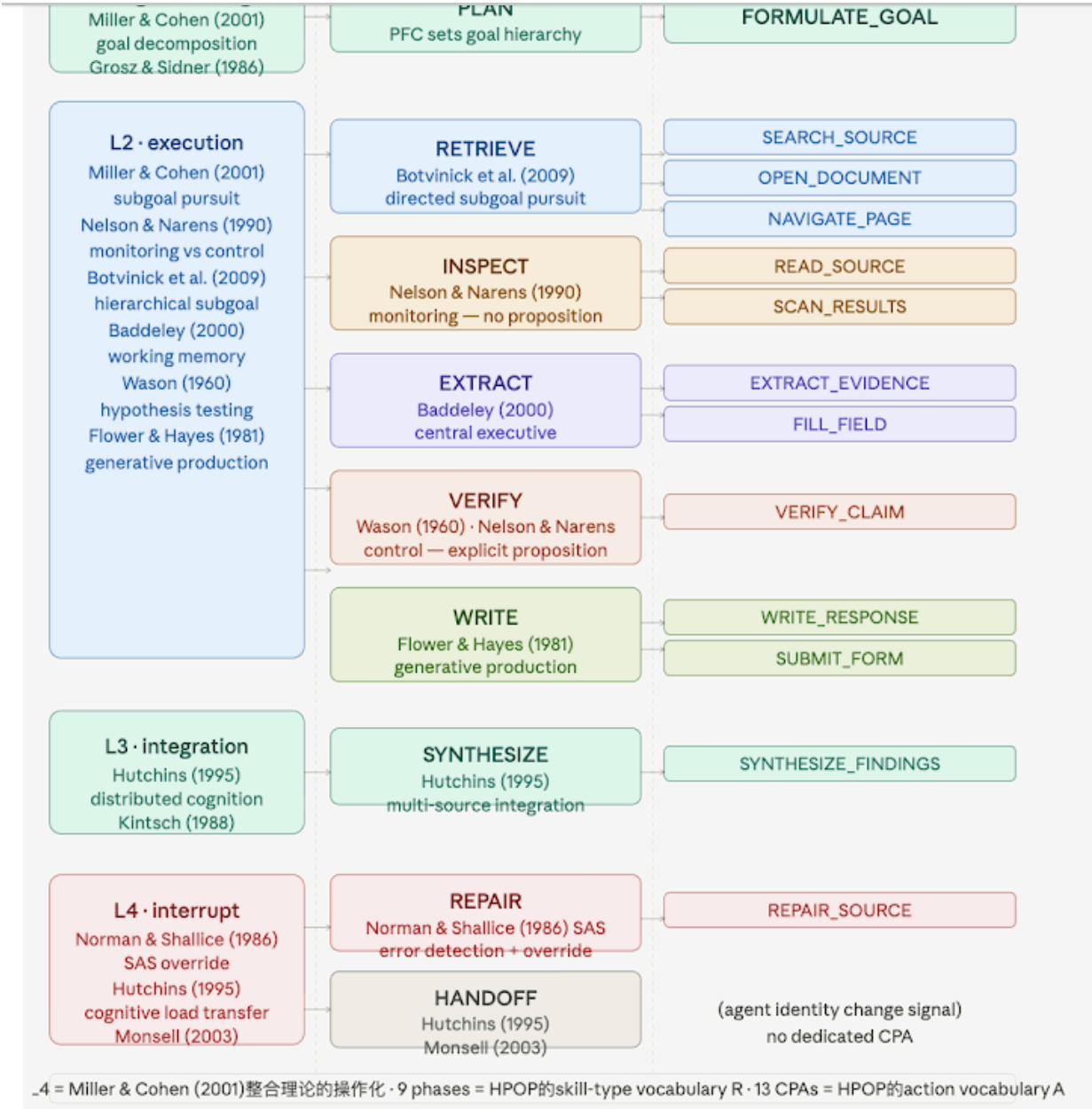
Script Theory	Schank & Abelson, 1977	skill	script	phase_sequence	script
		phase HPOP	trace HPOP	RETRIEVE formulation	"the same"
subroutine type may recur multiple times within a trace"					

Confidence Scoring / Human Review —

Cohen 1960 κ	quality baseline	Human Review	≥5	pattern→
active learning	Settles, 2009			

L1-L4 phase	9 CPA	phase CPA	CPA phase	9 phase
L1-L4 ↓	9 phase	HPOP	skill	R vocabulary
↓	13 CPA	HPOP	phase	A vocabulary
HPOP	vocabulary		segment boundary	partial order

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L1-L4

L1-L4 HPOP " PLAN REPAIR 9 phase phase"

Miller & Cohen 2001 9 phase L1-L4

9 phase HPOP skill-type vocabulary R

v3 9 phase HPOP R = {1,...,R} HPOP
segmentation segment phase label phase trace
T5 RETRIEVE

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CPA phase CPA Activity Theory Action — Operation click/scroll
Activity skill CPA physical actions HPOP
action vocabulary $A = \{1, \dots, M\}$ procedural UI

HANDOFF CPA

HANDOFF CPA

HANDOFF agent identity change agent_id agent_1 agent_2
Step 1a Physical Action Normalization action token agent_id Step
1b CPA HANDOFF physical actions
 procedural action